

**CLASSIFICATION OF ELEMENTS
AND PERIODICITY IN PROPERTIES**

- 61.** The element with positive electron gain enthalpy is

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- (a) hydrogen
- (b) sodium
- (c) oxygen
- (d) neon

- 62.** Consider the following statements:

- (i) The radius of an anion is larger than that of the parent atom.
 - (ii) The ionisation energy generally increases with increasing atomic number in a period.
 - (iii) The electronegativity of elements increases on moving down a group.
- Which of the above statements is/are correct?

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- (a) (i) alone
- (b) (ii) alone
- (c) (i) and (ii)
- (d) (ii) and (iii)

- 63.** Which of the following metals requires radiation of the highest frequency to cause the emission of electrons?

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- (a) Na
- (b) Mg
- (c) K
- (d) Ca

- 64.** Which of the following represents the correct order of increasing electron gain enthalpy with negative sign for the elements O, S, F and Cl?

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- (a) $\text{Cl} < \text{F} < \text{O} < \text{S}$
- (b) $\text{O} < \text{S} < \text{F} < \text{Cl}$
- (c) $\text{F} < \text{S} < \text{O} < \text{Cl}$
- (d) $\text{S} < \text{O} < \text{Cl} < \text{F}$

- 65.** An element A of group 1 shows similarity to an element B belonging to group 2. If A has maximum hydration enthalpy in group 1, then B is

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- (a) Mg
- (b) Be
- (c) Ca
- (d) Sr



66. Which is the correct order of electronegativity?

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- (a) $F > O > N > C$
- (b) $F > N > O > C$
- (c) $C < N < O < F$
- (d) $C > N > O > F$

67. The correct order of decreasing electronegativity values among the elements I—beryllium, II—oxygen, III—nitrogen and IV—magnesium is

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- (a) $II > III > I > IV$
- (b) $III > IV > II > I$
- (c) $I > II > III > IV$
- (d) $I > II > IV > III$

68. Which of the following elements is considered a metalloid?

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- (a) Sc
- (b) Pb
- (c) Bi
- (d) Te

69. The elements with zero electron affinity are

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- (a) Boron and Carbon
- (b) Beryllium and Helium
- (c) Lithium and Sodium
- (d) Fluorine and Chlorine

70. Which is not the correct order for the stated property?

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- (a) $Ba > Sr > Mg$; atomic radius
- (b) $F > O > N$; first ionisation enthalpy
- (c) $Cl > F > I$; electron affinity
- (d) $O > Se > Te$; electronegativity

71. The compounds of the s-block elements, with the exception of lithium and ...X..., are predominantly ionic. Here, X refers to

- (a) hydrogen
- (b) helium
- (c) magnesium
- (d) beryllium

72. Alkali metals are powerful reducing agents because

- (a) these are metals
- (b) their ionic radii are large
- (c) these are monovalent
- (d) their ionisation potential is low



73. An element having electronic configuration $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1$ forms

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- (a) acidic oxide
- (b) basic oxide
- (c) amphoteric oxide
- (d) neutral oxide

74. Which of the following sequences correctly represents the decreasing acidic nature of oxides?

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- (a) $\text{Li}_2\text{O} > \text{BeO} > \text{B}_2\text{O}_3 > \text{CO}_2 > \text{N}_2\text{O}_3$
- (b) $\text{N}_2\text{O}_3 > \text{CO}_2 > \text{B}_2\text{O}_3 > \text{BeO} > \text{Li}_2\text{O}$
- (c) $\text{CO}_2 > \text{N}_2\text{O}_3 > \text{B}_2\text{O}_3 > \text{BeO} > \text{Li}_2\text{O}$
- (d) $\text{B}_2\text{O}_3 > \text{CO}_2 > \text{N}_2\text{O}_3 > \text{Li}_2\text{O} > \text{BeO}$

75. Which property of an element is directly related to electronegativity?

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- (a) Atomic radius
- (b) Ionisation enthalpy
- (c) Non-metallic character
- (d) None of these

